

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Odd Semester)

Innovative Practices Description

UNIT I (BIASING OF DISCRETE BJT, JFET AND MOSFET)

Degree, Semester & Branch: V Semester B.E. ECEA

Course Code & Title: EC6501 Digital Communication

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Sampling & Quantization

Name of the Innovative Practice: Application Oriented Teaching

Description:

Sampling converts a time-varying voltage signal into a discrete-time signal, a sequence of real numbers. Quantization replaces each real number with an approximation from a finite set of discrete values. Quantization is the process where the least significant bit (LSB) is determined if the input analog voltage lies in the lowest sub-range of the input voltage range. The input analog voltage range from 251 mV to 500 mV will be assigned the digital code 001 and so on.

Quantization Noise:

It is a type of quantization error, which usually occurs in analog audio signal, while quantizing it to digital. For example, in music, the signals keep changing continuously, where regularity is not found in errors. Such errors create a wideband noise called as Quantization Noise.

The students were told about the concepts of sampling and quantization and its needs in digital communication. Lecture video also was played to the students.

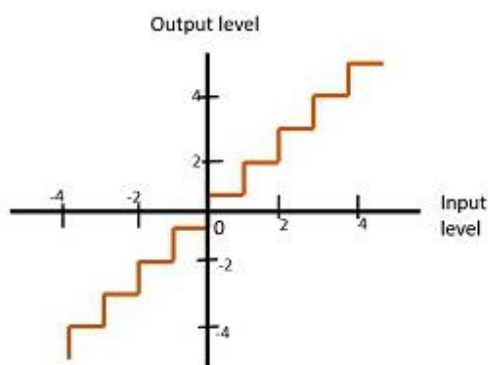


Fig 1 : Mid-Rise type Uniform Quantization

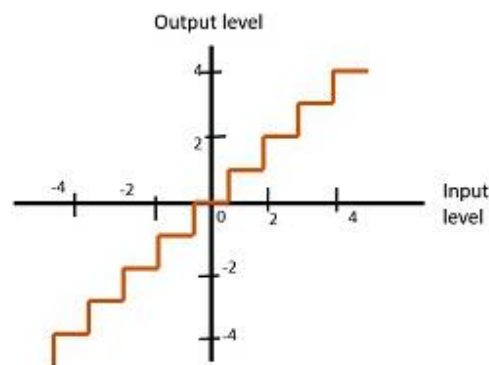


Fig 2 : Mid-Tread type Uniform Quantization

References:

1. <https://microchipdeveloper.com/adc:adc-quantization-error>
2. S.Haykin, "Digital Communications", John Wiley, 2005
3. J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001.